Performance-Monitoring Thresholds

A flag is triggered when CMJ jump height or RSI decreases by 10% or more below the athlete's rolling baseline. CMJ jump height reflects lower-body power, while RSI captures how quickly an athlete can generate force through the stretch-shortening cycle. Consistent evidence shows that reductions in either metric are strongly associated with neuromuscular fatigue and accumulated physical stress (Suarez-Arrones et al., 2020; Rabelo et al., 2022). Anićic et al. (2023) further demonstrated that force-plate jump variables show excellent reliability, with intraclass correlation coefficients frequently exceeding 0.90, indicating minimal normal day-to-day variability. Given this high measurement stability, a decline of 10% or greater exceeds expected noise and represents a meaningful reduction in neuromuscular performance. As a result, CMJ-based thresholds are widely used in applied sport science to monitor readiness, identify fatigue, and detect early signs of overload. In the Python implementation, any test in which CMJ or RSI drops by ≥10% is automatically flagged.

A flag is generated when the Accumulated Acceleration Load (AAL) for a session exceeds 150% of the athlete's rolling weekly average. Acceleration-based workload metrics capture mechanical stress from accelerations, decelerations, and rapid changes of direction that are not reflected by total distance alone (Bredt et al., 2020). Research indicates that abrupt increases in acceleration load are linked to an elevated risk of soft-tissue injury, particularly in field and court sports (Delves et al., 2021; Xiao et al., 2021). Because AAL represents "true" mechanical load, unusually large spikes may signal excessive strain or inadequate recovery between sessions. Setting the threshold at 1.5 times the athlete's rolling weekly mean aligns with red-zone criteria commonly used in acute-to-chronic workload models and helps identify sessions with acceleration demands that meaningfully deviate from normal patterns. This cutoff is supported by evidence showing that sudden workload spikes, rather than total volume alone, are more strongly associated with injury risk.

Finally, a flag is applied when an athlete's daily or session total distance falls 20% below the team- and position-specific median, indicating a potential deviation from expected movement demands relative to peers in similar roles.

Metric	Threshold Rule	Performance Meaning	Literature Support
Jump Height / RSI	≥10% drop from baseline	Neuromuscular fatigue	Anicic 2023; Rabelo 2022; Fitzpatrick 2021
AAL (Acceleration Load)	>150% of weekly average	Acute load spike (injury risk)	Delves 2021; Freitas 2025; Xiao 2021
Total Distance	<80% of position norm	Unusually low workload/participation	Benson 2020; Mallo 2015; Kupperman 2020

Evidence-Based Literature Justification

This performance-monitoring framework applies evidence-based thresholds across neuromuscular, mechanical, and volume-based metrics to identify meaningful changes in athlete readiness. A decline of 10 percent or more in countermovement jump (CMJ) height or reactive strength index (RSI) is flagged because both metrics are highly reliable indicators of lower-body power and stretch-shortening cycle function. Force-plate CMJ variables consistently demonstrate ICC values above 0.90, which indicates minimal natural variation and supports the interpretation of a 10 percent drop as a meaningful reduction in neuromuscular performance (Anicic et al., 2023). Reductions of this magnitude are associated with neuromuscular fatigue, decreased explosive ability, and accumulated training stress according to existing research (Suarez-Arrones et al., 2020; Rabelo et al., 2022).

Accumulated Acceleration Load (AAL) is flagged when a session exceeds 150 percent of an athlete's rolling weekly average. This threshold reflects findings that acceleration-based metrics capture high-force mechanical stresses such as accelerations, decelerations, and frequent directional changes that distance alone cannot quantify (Bredt et al., 2020). Sudden increases in acceleration load have been repeatedly linked to higher soft-tissue injury risk in field and court sports (Delves et al., 2021; Xiao et al., 2021). The 1.5 times cutoff is consistent with acute to chronic workload guidelines used to identify periods of elevated injury risk, although future work should examine how more conservative or more permissive thresholds influence detection accuracy.

Total distance is flagged when it falls below 80 percent of team and position-specific medians. Positional demands create predictable differences in movement volume, and modern tracking technologies such as GPS and local positioning systems reliably quantify distance at high sampling rates (Blauberger et al., 2021). Distances that fall substantially below positional norms can indicate reduced participation, restricted work rate, under-recovery, or early signs of fatigue (Mallo et al., 2015; Kupperman and Hertel, 2020).

Collectively, these thresholds provide a practical and individualized method for identifying neuromuscular fatigue, acute load spikes, and abnormal reductions in training volume. When integrated into a unified monitoring framework, they help practitioners contextualize athlete readiness, guide appropriate load adjustments, and detect early signs of overload using metrics that are validated and widely supported in the sports performance literature.

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